Database Management Systems – Practical Assignment 2

1) Install SQL Server DBMS and create the database

Create a database with the following schema:

(Requirement: Capture screenshots of the schema and the data instances of all tables in the database)

Given the following relational schemas for a computer manufacturing management database:

- **Manufacturer** (MaNSX, TenNSX, DiaChi)
- **Product** (MaNSX, MaSP, loai)
- **PC** (MaSP-P, CPU, RAM, HD, price)
- **Laptop** (MaSP-L, CPU, RAM, HD, Screen, price)

2) Practice tasks and submit a report *(File name: StudentID_StudentName.pdf)*

A) Use SQL to perform the following tasks

(Requirement: Write SQL queries, explain your query, and provide screenshots of the results)

1. Find the names of manufacturers that produce only PCs but not Laptops.
2. Increase the price of all PCs produced by **Dell** by 10%.
3. Find manufacturers that have at least two computers of different types (PC and Laptop) with the same CPU speed.
4. Find manufacturers of all products (PC or Laptop) with hard disk capacity of at least **256 GB**.
5. Find manufacturers of Laptops with the largest hard disk capacity.
6. Delete all manufacturers that do not produce any Laptop products.

B) Query Optimization

Rewrite and optimize the SQL queries for tasks 1–6 above.

(Requirement: Write SQL queries, explain your query, and provide screenshots of the results)

3) Database Application Development

Develop a database application with the following functions:

- Insert, update, delete data
- Search data
- Execute advanced (or optimized) queries

4) Oral Examination